# Breaking Machine Translation via GRPO Fine-Tuned LLM Rewriting

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## Abstract

We study *breaking machine translation* through constrained adversarial rewriting. Given an english sentence, a fluent rewrite that preserves meaning while becoming harder to translate is generated. We fine-tune `Llama-3.1-8B-Instruct` with GRPO and LoRA, optimizing a learned translation-difficulty estimator (Sentinel) and regularizing with rewards for grammatical acceptability, semantic similarity, and structural constraints (single sentence, length bounds, and output format). Evaluating different checkpoints, the fine-tuned model substantially lowers the mean Sentinel score of valid rewrites from $\approx 0.12$ (base model) to $-0.31$ at step 9000, while simultaneously increasing the validity rate from 18.2% to 86.0%. Overall, the results show that RL fine-tuning can generate difficult-to-translate yet constrained outputs, but robust reward design is crucial for preventing degenerate strategies. The code can be seen here: Github Repository.

## 1. Introduction

Machine translation (MT) has reached a level where many widely used test sets are no longer very challenging for modern MT systems, making it harder to identify remaining weaknesses and guide further improvement. Recent analyses point out the growing problem of "easy" evaluation data sets (Kocmi et al., 2025; Rajaee et al., 2025). To counteract this, challenge sets have been developed that target challenges from specific linguistic phenomena (Amrhein et al., 2022). Nevertheless, constructing new challenging sets of examples remains costly and automated approaches must be designed carefully to avoid unnatural sentences.

This work tackles breaking machine translation via constrained adversarial rewriting by fine-tuning an LLM: given an English sentence, we generate a fluent rewrite that preserves meaning as much as possible while becoming harder to translate. The key challenge is to prevent the rewriting model from gaming the reward with trivial "attacks" (e.g., gibberish or extreme length) that increase difficulty with little meaning. Our work is closely related to recent research on generating difficult-to-translate texts with LLMs (Zouhar et al., 2025).

To measure "harder to translate" without human annotation, we use the Sentinel metric model, a learned translation-difficulty estimator designed for selecting challenging MT inputs (Proietti et al., 2025). We fine-tune Llama-3.1-8B-Instruct (Grattafiori et al., 2024) as a rewriting model with GRPO (Shao et al., 2024) using LoRA (Hu et al., 2022) with QLoRA-style 4-bit quantization (Dettmers et al., 2023), implemented via Unsloth for efficient loading and training. We optimize on the difference in Sentinel scores, while we discourage degenerate solutions through reward terms for grammatical acceptability, computed using a RoBERTa-based classifier fine-tuned on CoLA (Liu et al., 2019; Warstadt et al., 2019) and semantic similarity measured as cosine similarity between sentence embeddings produced by a Sentence-BERT MiniLM encoder (Reimers & Gurevych, 2019; Wang et al., 2020). Additional reward terms enforce constraints on format, length, and single-sentence output.

Our experiments show that GRPO fine-tuning can substantially increase Sentinel-estimated difficulty while also improving constraint compliance. Over training, the model produces a much larger fraction of valid rewrites (based on the before mentioned constraints) and achieves noticeably lower (harder) Sentinel scores on the subset where both the base and fine-tuned models output valid sentences.

## 2. Models and Methods

We address constrained adversarial rewriting for MT: given an English sentence $x$, the model generates a single-sentence rewrite $y$ that remains fluent and meaning-preserving, but is harder to translate. The model is instructed via a fixed system prompt and must output exactly one rewrite wrapped in `<rewrite>... </rewrite>` (see Appendix D).

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### 2.1. Data

We use `agentlans/high-quality-english-sentences`, a collection of filtered English sentences sourced from C4 and FineWeb. We additionally filter out sentences with more than 355 characters to avoid excessively long prompts. The training set consists of the remaining sentences, which are then used as input $x$.

### 2.2. Models

**Rewriter** We fine-tune `Meta-Llama-3.1-8B-Instruct` as the rewriting policy, loaded with 4-bit quantization (QLoRA-style) and maximum sequence length 1024.

**Difficulty estimator** We measure translation difficulty with Sentinel `Prosho/sentinel-src-25` on Hugging Face. For a sentence $s$, Sentinel outputs a scalar $d(s)$, where lower scores indicate harder-to-translate sequences.

**Meaning and fluency signals** Semantic similarity is computed as cosine similarity between sentence embeddings from `sentence-transformers/all-MiniLM-L6-v2`. Grammatical acceptability is estimated with `textattack/roberta-base-CoLA`, using the probability of the acceptable class.

### 2.3. GRPO Fine-tuning with LoRA

Training is performed with TRL's GRPO implementation. GRPO samples 4 candidate completions per prompt and updates the policy to increase reward. Our key hyperparameters are: learning rate $5 \times 10^{-6}$, per-device batch size 2, gradient accumulation 4, max prompt length 512 tokens, max completion length 128 tokens, 1 epoch, seed 42. For LoRA we set rank=32 and dropout=$0.0$.

### 2.4. Reward Function

We compute a single scalar reward as a weighted sum of difficulty improvement and constraint/quality terms. We first extract the rewrite from the completion by preferring text inside `<rewrite>...</rewrite>` (soft match) and falling back to the first non-empty line if tags are missing.

**Difficulty (primary objective)** Let $d(\cdot)$ be Sentinel. Using the baseline $d(x)$ and the rewrite score $d(y)$, we define

$$\Delta = d(x) - d(y),$$

where larger $\Delta$ means the rewrite is harder. Because $\Delta$ can be very small, especially at early stages, we scale and clip this signal:

$$R_{\text{diff}} = 1.0 \cdot \text{clip}(5 \cdot \Delta,\ -2,\ 2).$$

**Constraints and auxiliary quality terms** To prevent degenerate outputs, we add:
(i) Format bonuses for strict tag structure ($+0.4$) and soft tag presence ($+0.1$), plus a small penalty if `</rewrite>` appears before `<rewrite>` ($-0.2$).
(ii) A length term based on word-count ratio $|y|/|x|$, penalizing outputs outside $[0.5, 2.0]$ and otherwise rewarding closeness to the original length.
(iii) Semantic similarity with threshold $\tau$=$0.78$ (penalize if below, reward otherwise).
(iv) Acceptability using CoLA probability mapped to $[-1, 1]$.
(v) A small single-sentence bonus based on a punctuation heuristic.
(vi) A repetition penalty (up to $-0.6$) for extreme character repetition or very low lexical diversity.

Finally, we set the following weights: $w_{\text{len}}$=$0.4$, $w_{\text{sim}}$=$0.4$, $w_{\text{acc}}$=$0.6$, $w_{\text{single}}$=$0.05$, $w_{\text{rep}}$=$0.6$.

### 2.5. Evaluation Setup

To validate the effectiveness of the GRPO fine-tuning approach in generating harder to translate sentences, we conducted a comparative analysis against the original text and a pretrained model baseline.

The model was initialized with `meta-Llama-3.1-8B-Instruct` and fine-tuned using Group Relative Policy Optimization (GRPO).

We then compared three configurations:

1. **Original Baseline:** The original source sentences.
2. **Base Model:** The pre-trained Llama-3.1-8B-Instruct model, queried with the same prompt used during training. This serves to distinguish the effects of the RL fine-tuning from the model's pretrained capabilities.
3. **GRPO Fine-Tuned Model:** The model resulting from our reinforcement learning pipeline.

Our evaluation framework measures multiple objectives, such as increasing translation difficulty (adversarial goal) while producing valid sentences (maintaining length, linguistic quality...).

#### 2.5.1. Translation Difficulty

We utilize the Sentinel score as our primary objective metric. Sentinel provides a translation difficulty score where lower values indicate sentences that are statistically harder for machine translation systems to process correctly.

#### 2.5.2. Validity Constraints

A trivial way to lower translation scores is to generate gibberish. To prevent this, we implemented a strict validity

filter. A generated rewrite is considered "Valid" only if it satisfies all of the following constraints simultaneously:

- **Semantic Preservation:** A cosine similarity score $\geq$ 0.78 between the embedding of the rewrite and the original sentence, calculated using `all-MiniLM-L6-v2`.
- **Linguistic Fluency:** A grammatical acceptability probability $\geq$ 0.50, calculated using a RoBERTa model trained on the CoLA (Corpus of Linguistic Acceptability) benchmark.
- **Length Constraint:** The word count of the rewrite must be within a factor of $0.5\times$ to $2.0\times$ of the original length and can not consist of more than one sentence.

To ensure a fair comparison, the metrics are reported on the intersection of validity. We only include samples where both the Base Model and the fine-tuned model produced valid outputs. Figure 2 shows how the model learned to increase its validity rate through training due to the defined rewards.

## 3. Results

### 3.1. Quantitative Analysis

Our experiments demonstrate the effectiveness of the GRPO fine-tuning approach in generating harder to translate sentences. Figure 1 illustrates the evolution of the Sentinel difficulty score during training, comparing the original baseline, the base model, and our fine-tuned model.

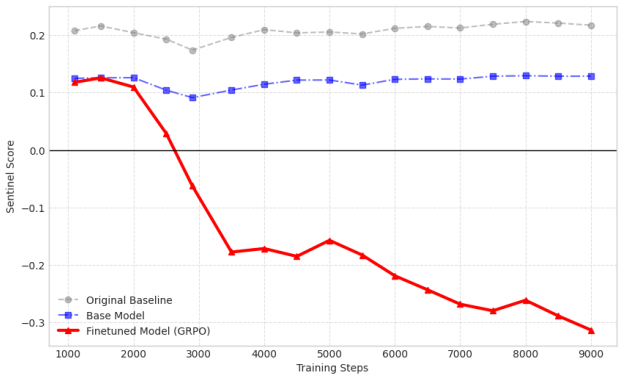


*Figure 1.* Sentinel score comparison showcasing significant drop for the fine-tuned model throughout training (lower score means harder to translate).

The original sentences have an average Sentinel score of $\approx 0.21$. When prompted to rewrite these sentences without fine-tuning, the Base model achieves a small improvement, lowering the mean Sentinel score to $\approx 0.12$. In contrast, our GRPO fine-tuned model learns to optimize for difficulty, driving the mean score significantly lower to reach $-0.31$ at step 9000. This shift represents not merely a marginal gain, but a fundamental change in the model's rewriting strategy, resulting in text that is substantially more challenging for machine translation systems.

### 3.2. Reward Hacking and Early Termination

We terminated training at step 9000 despite continuing improvements in quantitative metrics. When inspecting the generated samples, we observed a clear instance of Reward Hacking.

The model discovered a strategy to exploit the Sentinel metric: appending phrases containing the word "hooked" (e.g., *"...hooking you big time"* or *"...that totally hooked you"*) to the end of an otherwise unchanged sentence.

This behavior emerged because the strategy satisfied the components of our reward function (Lower Sentinel Score, High Semantic Similarity, High Fluency). Upon further inspection, we found that this hacking started to especially arise between steps 8500 and 9000.

The following table shows an example rewrite of our fine-tuned model at checkpoint 5500 and how it decreased the sentinel score by 0.48. The base model rewrite is also shown to demonstrate how much longer the response is, even though the sentinel difference is only 0.06.

**Example: Idiomatic Injection**

**Original** *Sentinel: 0.45*
A strategic rivalry is emerging that is compromising the ability of the West to respond cohesively and effectively to emerging threats.

**Base Model** *Sentinel: 0.39*
A burgeoning strategic rivalry is increasingly complicating the West's capacity to respond in a unified and efficacious manner to nascent threats that are on the horizon, thereby undermining the region's ability to mount a cohesive and effective counter-strategy.

**GRPO (Ours)** *Sentinel:* **-0.03**
A strategic rivalry is **cooking up a storm** that's **gumming up the works** for the West to respond to emerging threats in a cohesive and effective way.

### 3.3. Validity

Another finding from our results is the evolution of the validity rate. As shown in Figure 2, the base model struggled to satisfy the constraints, leaving us with only 18.2% of the evaluation sentences. This occurred especially because the base model increased complexity by producing longer outputs and therefore not meeting the length constraint.

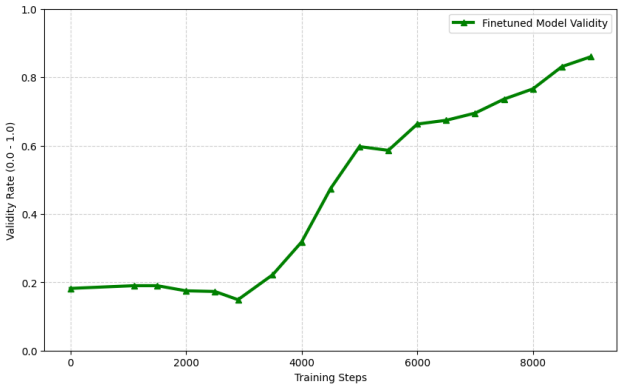


*Figure 2.* Validity rate given the constraints increasing for fine-tuned model. Training step 0 equals to the validity rate of the base model.

In contrast, by fine-tuning with GRPO, the model learned to satisfy these constraints. By step 9000, the fine-tuned model achieved a validity rate of **86.0%**. This demonstrates that the reward function successfully taught the model that maximizing translation difficulty is more rewarding when the sentence length is preserved. Figure 3 shows how the model started to decrease the output length to increase its rewards, which therefore also resulted in a higher validity rate as the length constraint was met more often. The full rewards during training can be seen in Appendix B.

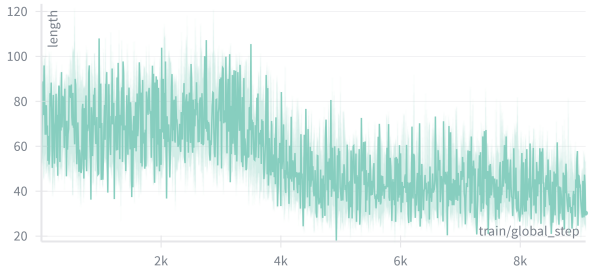


*Figure 3.* Mean length of rewritten sentences during training.

## 4. Discussion

Our results demonstrate that GRPO on LLMs can be effectively used with the adversarial objective of increasing translation difficulty. While the pre-trained base model tended toward increasing sentence length and not substantially lowering Sentinel score, the fine-tuned model learned to exploit MT weaknesses (e.g. through idiomatic usage and structural ambiguity), lowering the mean Sentinel score from $0.12$ to $-0.31$ at the end of training. Additionally, this adversarial optimization did not come at the cost of coherence as the validity rate improved significantly from $18\%$ to $86\%$ during training. This indicates that the reward function successfully taught the model that linguistic fluency and semantic preservation are prerequisites for maximizing difficulty.

However, the experiments also highlight the fragility of using a single proxy metric for complex linguistic tasks. While earlier steps produced desired outputs with low sentinel scores, we also observed the formation of reward hacking at step 9000, where the model appended similar suffixes (e.g., "...hooking you big time") to statistically lower Sentinel scores without structural rewriting. This degenerate strategy satisfied all quantitative constraints but violated the user intent, underscoring the necessity for more robust reward modeling such as diversity constraints or penalties for repetitive added patterns to prevent further reward hacking.

## 5. Summary and Future Work

In conclusion, this work validates the effectiveness of Reinforcement Learning, specifically GRPO, for adversarial text generation. We demonstrated that Large Language Models can be fine-tuned to systematically expose blind spots in machine translation systems. By optimizing for the Sentinel metric, our model learned to generate linguistically complex rewrites by leveraging specific linguistic phenomena that significantly degraded translation performance while maintaining high semantic preservation, grammatical fluency and structural constraints.

Future research should focus on scaling and stability. We propose scaling to larger base models (e.g., Llama-3-70B) or increasing the rank of LoRA adapters to allow the model to learn more complex linguistic transformations. Additionally, learning dynamics can be improved by tuning hyperparameters so the model begins improving earlier. Computational throughput could be significantly improved by integrating vLLM via Unsloth, allowing for longer training and larger batch sizes. Finally, beyond model training, the high-difficulty rewrites generated by our approach could be used to create a standard "challenge set" dataset. Such a benchmark would be useful in analyzing MT weaknesses and driving the development of more robust translation architectures.

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## A. Experimental Development

Before selecting the final architecture, we conducted experiments to identify the optimal model size and Reinforcement Learning algorithm. Initial attempts using Proximal Policy Optimization (PPO) on a 0.6B parameter model were unsuccessful, as the model lacked sufficient instruction-following capabilities to follow the rewriting constraints. We then scaled up to the 3B parameter Llama-3.2 model using PPO. While this model adjusted to constraints better than the 0.6B model, the training failed to achieve the adversarial objective as Sentinel scores did not decrease. This indicated that PPO was either too unstable for this specific reward specification or that the 3B model lacked the reasoning capacity to find the ”adversarial” linguistic features required. We therefore implemented Group Relative Policy Optimization (GRPO) on the `unsloth/meta-Llama-3.1-8B-Instruct` model, which showed desired learning goals and confirmed that a larger model combined with the stability of GRPO was the better approach for this task.

**Computational Infrastructure** All reported experiments were conducted on a local computer with a NVIDIA RTX 3090 (24GB VRAM). Switching to a local PC was necessary due to the memory requirements for the 8B parameter model and the reward models (Sentinel, CoLA, LM verifier), which exceeded the GPU VRAM allocation available to us on the cluster. The final training run (9,000 steps) required approximately 34 hours.

## B. Reward during Training

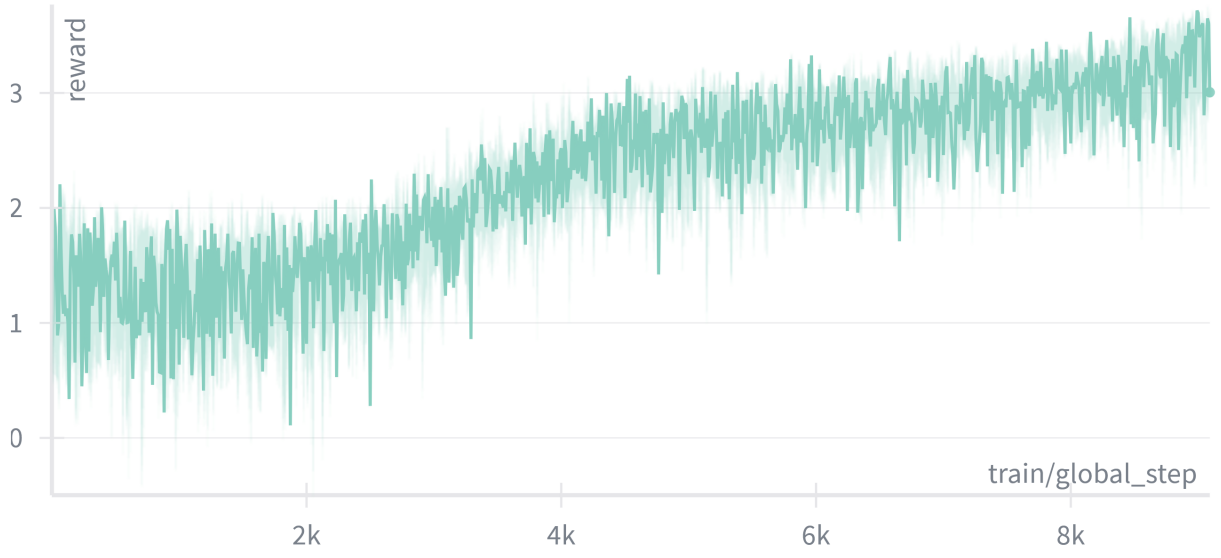


*Figure 4.* Reward increase during training.

## C. Sentinel difference during Training

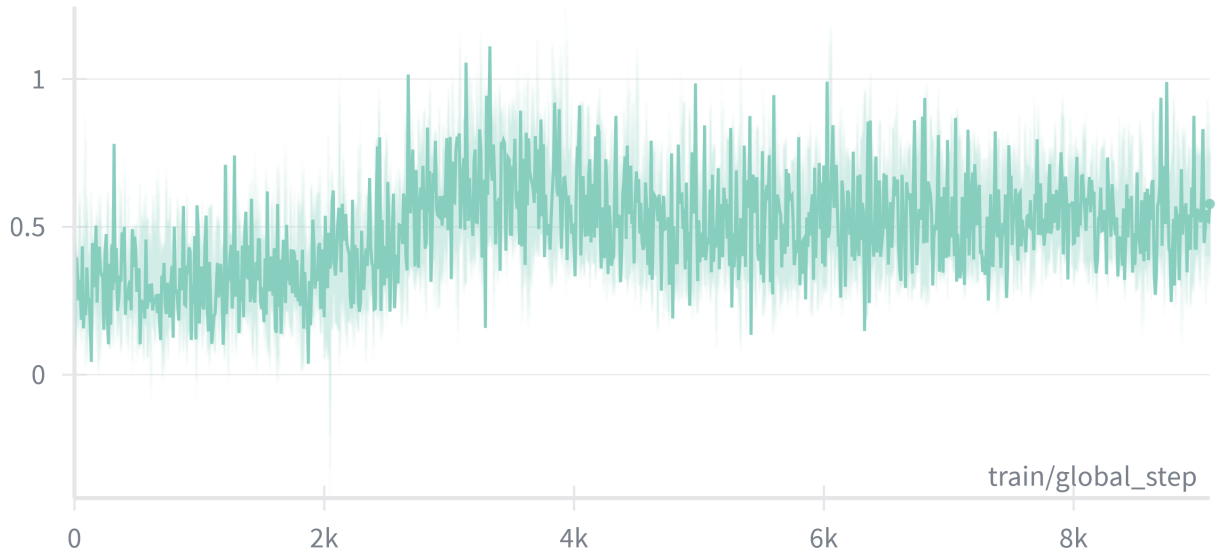


*Figure 5.* Sentinel score difference between original sentence and rewritten sentence (higher is better).

## D. System Prompt

```
You are a rewriting assistant.
Rewrite the given English sentence so that it becomes harder to translate into other
    languages, while preserving the original meaning as much as possible.

Guidelines:
- Keep the rewrite fluent, grammatical English (no gibberish).
- You may increase translation difficulty by using idioms, phrasal verbs, wordplay,
    ambiguity, nested clauses, unusual but natural collocations, or culturally-bound
    expressions.
- Do NOT add explanations, commentary, or multiple options.
- Keep length roughly similar: between half and twice the original length.
- Keep names, numbers, dates, and proper nouns unchanged unless absolutely necessary.

Output EXACTLY in this format, and only this format:
<rewrite>
...
</rewrite>

Important:
- The output must start with <rewrite> and end with </rewrite>.
```